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Unit 4 : Session 10 : SLO 1 | Odd Man Out Pipelining Performance

I. Odd Man Out

1. Group CPI, IPC, Speedup, Bandwidth → Bandwidth → a data-transfer rate; the others are instruction-level performance metrics.
2. Group Throughput, Latency, Clock rate, Hard disk RPM → Hard disk RPM → a storage-device spec, not a CPU pipeline metric.
3. Group Ideal speedup, Observed speedup, Pipeline depth, Cache size → Cache size → a memory-hierarchy parameter, not a pipeline speedup/depth measure.
4. Group Data forwarding, Branch prediction, Hazard elimination, Motherboard → Motherboard → a physical board; the others are pipeline hazard techniques.
5. Group Structural, Control, Data hazard, Power outage → Power outage → an external failure, not a pipeline hazard.
6. Group Superpipelining, Superscalar, VLIW, Solid-state drive → Solid-state drive → storage; the others are ILP architectures.
7. Group Bubble, Flush, Stall, Acceleration → Pipeline acceleration → the others are pipeline disruption terms.
8. Group ILP, Thread-level parallelism, Sequential processing → Sequential processing → no parallelism.
9. Group 1 CPI, 0.5 CPI, 2 CPI, 5 GHz → 5 GHz → a clock frequency; the others are cycles-per-instruction values.
10. Group RISC pipeline, CISC pipeline, Printer driver → Printer driver → peripheral software, not a processor pipeline.